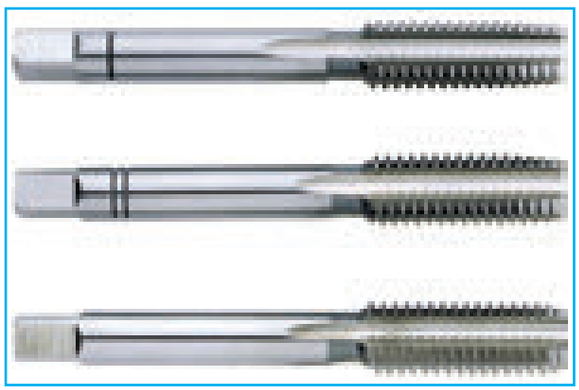
Tap

A tap has threads like a bolt. It has three or four flutes cut across the threads. It can cut threads on the inside of a hole. The flutes on the threads from the cutting edges. It is a multi point cutting tool. It will dig into the walls of the hole as the lower past of the tap is slightly tapered. The shank of the tap is square shaped to enable it to be held by a tap wrench.

Taps are made up of carbon steel (or) high speed steel. Two types of taps are used. They are (i) hand tap and (ii) machine tap



Tap

2.16 Twist drill nomenclature

Twist drill nomenclature is shown in the given figure.

Axis

It is the longitudinal centreline of the drill running through the centres of the tang and the chisel edge.

Body

It is the part of the drill from its extreme point to the commencement of the neck. If present otherwise it is the part extending up to the commencement of the shank. Helical grooves are cut on the body of the drill.

Shank

It is part of the drill by which it is held and driven. It is found just above the body of the drill. The shank may be straight (or) taper. The shank of the drill can be fitted direct into the spindle (or) by a tool holding device.

Tang

The flattened end of the taper shank is known as tang. It is meant to fit into a slot in the spindle (or) socket. It ensures a positive drive of the drill.

Neck

It is the part of the drill which is diametrically undercut between the body and the shank of the drill. The size of the drill is marked on the neck.

Point

It is the sharpened end of the drill. It is shaped to produce lips. Faces, flanks and chisel edge.

Lip

It is the edge formed by the intersection of flank and face. There are two lips and both of them should be of equal length. Both lips should be at the same angle of inclination with the axis. (59^0).

Land

It is the cylindrically ground surface on the leading edges of the drill flutes adjacent to the body clearance surfaces. The alignment of the drill is maintained by the land. The hole is maintained straight and to the right size.

XII BME_English version CHAPTER 02.indd 46 1/8/2020 1:44:54 PM